

Deotgyeojang Meju R&D Fermentation Journal

"Our records become a new map for deotgyeojang meju."

This journal will serve as valuable research data for the R&D process of fermenting summer deotgyeojang meju with care over the course of one month. Please observe the meju at a set time each day or week, and record your honest impressions.

Participant Basic Information

Researcher (name of person completing this journal): _____

Region of residence: _____ City/County _____ District/Town/Village

What are the environmental characteristics of where you live? ☐ Inland/mountainous ☐ Urban/flatland ☐ Coastal area

Fermentation start date: 2026 _____ month _____ day

Meju shape, size, and count: (e.g., donut-shaped / round shape, total _____ pieces)

Heat treatment (roasting) method: ☐ Direct frying pan ☐ Oven / air fryer ☐ Other

Fermentation location: ☐ Indoor room ☐ Veranda ☐ Inside a heated fermentation chamber ☐ Other

Daily Observation Journal (Week 1: Intensive Monitoring)

※ The first week is the most critical period for microbial establishment and temperature control. Please observe at a consistent time each day.

[Basic Observation Item Guide]

Weight: Place the whole meju on a scale and record the amount of moisture evaporated.

Color stage: [1] Roasted brown [2] White microbial growth [3] Yellow / yellow-green mold growth

Aroma (smell): Roasted-barley smell, savory meju smell, sour smell, musty/pungent smell, sweet saccharification smell, etc.

Core / ambient temperature: Internal temperature inside the meju and the external room temperature where it is placed

■ [Day 1] Shaping, Roasting, and Fermentation Setup

Date/time: _____ month _____ day _____ o'clock | Ambient temp: _____ °C / Humidity: _____ %

Total meju weight: _____ g | Meju core (internal) temperature: _____ °C

Visible changes (color and appearance):

Surface condition: ☐ Moist ☐ Roasted and firm (confirms artificial surface-drying) ☐ Crispy

Meju color: _____

Changes detected by smell (aroma):

Notable observations / one-line memo: (e.g., Covered with cheesecloth, then closed a glass lid on top to retain moisture.)

■ [Day 2–Day 3] Full-Scale High-Temperature Fermentation & Bacillus Establishment

[Day 2] Weight: _____ g | Core temp: _____ °C / Humidity: _____ % | Condensation: Yes / No

Observations and aroma: _____

[Day 3] Weight: _____ g | Core temp: _____ °C / Humidity: _____ % | Condensation: Yes / No

Observations and aroma: _____

Notable issues (e.g., fruit fly infestation): _____

■ [Day 4–Day 7] Saccharification Promotion & Stabilization (End of Week 1)

[Day 4] Weight: _____ g | Core temp: _____ °C | Status: ☐ No change ☐ White mycelium formation begins

[Day 5] Weight: _____ g | Core temp: _____ °C | Status: _____

[Day 6] Weight: _____ g | Core temp: _____ °C | Status: _____

[Day 7] Weight: _____ g | Core temp: _____ °C | Status: ☐ White/yellow mold fully established ☐ Savory aroma intensifies

Weekly Observation Journal (Weeks 2–4: Long-Term Ripening & Full Maturation)

※ From Day 8 onward, the meju enters a period of self-ripening and drying. Record the changes weekly as they deepen.

■ [Week 2 Check] Fermentation Days 8–14 (Rapid Fermentation Complete, Entering Ripening)

Check date: _____ month _____ day (Fermentation day _____)

Total meju weight: _____ g (approx. _____ g lost from initial weight)

Core temp / humidity: _____ °C / _____ %

Appearance & microbial coat: ☐ White mycelium fully covers meju ☐ Ripened with yellow-green / yellow mold

Aroma changes: ☐ Roasted barley aroma ☐ Cheonggukjang (fermented soybean) aroma ☐ Deep koji/meju aroma ☐ Sweet rice-syrup (jocheong) aroma

My impressions and actions taken: (e.g., Turned off the fermentation chamber and moved the meju to a well-ventilated shaded spot to begin slow drying.)

■ [Week 3 Check] Fermentation Days 15–21 (Gentle Drying & Moisture Adjustment)

Check date: _____ month _____ day (Fermentation day _____)

Total meju weight: _____ g

Texture when touched: ☐ Still soft inside ☐ Firm outside, chewy inside ☐ Completely firm

Aroma character detected: _____

Notable observations / memo: _____

■ [Week 4: Final Maturation] Fermentation Days 22–30 (One-Month Completion, Just Before Breaking Apart)

Check date: _____ month _____ day (Fermentation day _____ – Final)

Final total meju weight: _____ g

Final meju color and appearance: _____

Breaking apart & cross-section observation:

What is the condition of the interior when the meju is broken open? (e.g., Fermented evenly to a golden-yellow color throughout; moist at the core, etc.)

Color and mold distribution of the broken cross-section: _____

Overall R&D Reflection (Concluding the One-Month Journey)

If you were to describe the final “aroma” of your deotgyeojang meju, fermented over the month, in three words, what would they be?

[], [], []

During the 30-day fermentation, what was the most challenging moment, or the point requiring the most caution?

(e.g., Could not control humidity in Week 2, causing excess surface moisture; lack of airflow during final drying caused a musty smell, etc.)

What is your final one-line reflection on taking part in this R&D project to carefully cultivate “summer deotgyeojang meju” over the course of a month?

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